Module-Wrestler Online Character Dictionary

Gabe Field

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Abstract

We are the most modern parody of a long-established dictionary. We relate the language of characters with the language of finite-dimensional complex group modules. The main meat of this document is in subsection 1.3 where we draw direct parallels between properties of groups or their representations to properties of the characters, and in subsection 1.2 where we collect some facts of the characters themselves. A fiction of the story of characters is provided (mostly for fun, though I hope it builds an appreciation for why characters should be useful and natural). All proofs and some character tables are included.

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1 The Translation Dictionary

There are categories **FinGroup** of finite groups, **FinVect** of finite-dimensional vector spaces over $\mathbb{C}$, and for each $G : \mathbf{FinGroup}$, a category $\mathbf{FinVect}^G$ of finite-dimensional complex G-modules. If $V : \mathbf{FinVect}^G$, the G-action will be denoted $G \xrightarrow{\rho_V} \mathrm{GL}(V)$. The set of conjugacy classes will be denoted $\mathcal{CC}(G)$. For $x \in G$, its conjugacy class is denoted $\mathcal{CL}(x)$.

1.1 Definitions in the Language of Characters

Throughout, $G : \mathbf{FinGroup}$ is a group and $V : \mathbf{FinVect}^G$ is a finite-dimensional complex G-module. (Any other type data will be specified.)

(D1) **Character** of V: The function $\chi_V := \mathrm{tr} \circ \rho_V : G \to \mathbb{C}$. It is a class function (D3), so it is often factored through the quotient $G \twoheadrightarrow \mathcal{CC}(G)$, where it is regarded as a map $\mathcal{CC}(G) \to \mathbb{C}$.

(D2) **Character table** of G: A listing of the characters χ_I corresponding to the finite-dimensional complex irreducible modules I of G. As characters are class functions (D3), the value on each *conjugacy class* is listed. As the irreducible modules biject with the conjugacy classes, this is a square matrix. For example, behold the character table of S_2 (it's not interesting):

Character	$(*)(*)$	$(**)$
χ_{trivial}	1	1
χ_{sign}	1	-1

Beware – although the row for χ_I encodes the character χ_I, it is missing information on the size of the conjugacy class. This information is critical to computing the inner product of two characters (D5); generically, the inner product **does not** agree with the "dot product".

(D3) **Class function**: A function $G \to A$ which is constant on conjugacy classes of G. A is typically $\mathbb{C}$. These form a vector space $\mathbb{C}^{\mathcal{CC}(G)}$.

(D4) **Kernel** of the character χ_V: The normal subgroup $\{g \in G \mid \chi_V(g) = \chi_V(1)\}$. More details at (T12).

(D5) **Inner product** on $\mathbb{C}^G$: The vector space of functions $G \to \mathbb{C}$ carries Hermitian inner product $\langle -, - \rangle$ given by

$$\langle \varphi, \psi \rangle := \frac{1}{|G|} \sum_{g \in G} \varphi(g) \cdot \overline{\psi(g)}$$

or, for characters $(U, V : \mathbf{FinVect}^G)$:

$$\langle \chi_U, \chi_V \rangle := \frac{1}{|G|} \sum_{g \in G} \chi_U(g) \cdot \chi_V(g^{-1})$$

because of (G3). Restricting to the linear subspace $\mathbb{C}^{\mathcal{CC}(G)}$ gives an inner product on the class functions.

1.2 Grammatical Rules of the Language of Characters

I mean in this section to list some key properties that characters always satisfy. If you were to communicate using characters, these are some rules that "always appear" in your sentences! Throughout, $G : \mathbf{FinGroup}$ and $U, V : \mathbf{FinVect}^G$ unless otherwise specified. The set of conjugacy classes is denoted $\mathcal{CC}(G)$, and $\mathcal{I}(G)$ denotes the set of irreducible finite-dimensional complex G-modules, listed once per isomorphism class. Conjugacy classes are denoted $\mathcal{CL}(g)$ and centralisers by $\mathcal{CN}(g)$.

(G1) **Column orthogonality of character tables**: Listing a character table with each character along the rows and each conjugacy class along the columns,

$$X := \begin{array}{c} \\ \chi_{\text{trivial}} \\ \chi_{I_1} \\ \vdots \\ \chi_{I_{k-1}} \end{array} \begin{array}{c} \begin{array}{cccc} \mathcal{CL}(1_G) & \mathcal{CL}(g_1) & \cdots & \mathcal{CL}(g_{k-1}) \end{array} \\ \begin{pmatrix} * & * & \cdots & * \\ * & * & \cdots & * \\ \vdots & \vdots & \ddots & \vdots \\ * & * & \cdots & * \end{pmatrix} \end{array}$$

we have that each column is orthogonal under the usual dot product[1]. That is, for $g, h \in G$ with distinct conjugacy classes $\mathcal{CL}(g) \neq \mathcal{CL}(h)$,

$$\sum_{i=0}^{k-1} \chi_{I_i}(g)\overline{\chi_{I_i}(h)} = 0$$

whilst to give the square-norm of a column, we have

$$\sum_{i=0}^{k-1} |\chi_{I_i}(g)|^2 = \frac{|G|}{|\mathcal{CL}(g)|} = |\mathcal{CN}(G)|.$$

(G2) **Dimensions/degrees of irreducibles divide order of group**: For $I \in \mathcal{I}(G)$,

$$\dim(I) = \chi_I(1_G) \mid |G|.$$

(G3) **Inverses**: For $g \in G$, $\chi_V(g^{-1}) = \overline{\chi_V(g)}$.

(G4) **Frobenius reciprocity**: If $H \leq G$, $U : \mathbf{FinVect}^H$ and $V : \mathbf{FinVect}^G$, then

$$\langle \operatorname{ind}_H^G \chi_U, V \rangle = \langle U, \operatorname{res}_H^G \chi_V \rangle.$$

See (T2) and 2.9.

(G5) **Orthonormality of irreducibles**:

$$U, V \text{ both irreducible} \implies \langle \chi_U, \chi_V \rangle = \mathbf{1}_{U \simeq V} \quad \text{(indicator function)}.$$

(G6) **Orthonormal basis formed by irreducibles**: The vector space $\mathbb{C}^{\mathcal{CC}(G)}$ carries orthonormal basis $\{\chi_I \mid I \in \mathcal{I}(G)\}$. The inner product is **not** the "dot product", but rather as defined at (D5).

1.3 Translating Between Characters and Representations

Throughout, $G : \mathbf{FinGroup}$ and $U, V : \mathbf{FinVect}^G$.

(T1) **Abelian**: The group G is Abelian just when its character table is $|G| \times |G|$, itself just when each irreducible character is one-dimensional.

(T2) **Adjunction of induction to restriction**: For a subgroup $H \leq G$, the adjunction of functors $\operatorname{Ind}_H^G \dashv \operatorname{Res}_H^G$ appears as **Frobenius reciprocity**:

$$\mathbf{FinVect}^G(\operatorname{Ind}_H^G U, V) \simeq_{\text{nat.}U,V} \mathbf{FinVect}^H(U, \operatorname{Res}_H^G V) \rightsquigarrow \langle \operatorname{ind}_H^G \chi_U, \chi_V \rangle = \langle \chi_U, \operatorname{res}_H^G \chi_V \rangle.$$

Here, $\operatorname{ind}_H^G \chi_U := \chi_{\operatorname{Ind}_H^G U}$ and $\operatorname{res}_H^G \chi_V := \chi_{\operatorname{Res}_H^G \chi_V}$; extending linearly, ind_H^G and res_H^G define linear maps $\operatorname{ind}_H^G : \mathbb{C}^{\mathcal{CC}(H)} \rightleftarrows \mathbb{C}^{\mathcal{CC}(G)} : \operatorname{res}_H^G$. The inner product relation is a decategorification of the (categorical) adjunction.

[1] This inner product is *not* weighted by the sizes of the conjugacy classes.

(T3) **Class number**: The number of conjugacy classes of G is the same as $|\mathcal{I}(G)|$.

(T4) **Commutator subgroup**: The commutator subgroup $[G, G]$ is the intersection of all one-dimensional characters.

$$[G, G] = \bigcap_{I \in \mathcal{I}(G).\ \dim I = 1} \operatorname{Ker} \chi_I.$$

(T5) **Dimension/Degree** of V:

$$\dim(V) = \chi_V(1_G).$$

(T6) **Dimension of hom-space** $\mathbf{FinVect}^G(U, V)$ (this space often goes by $\operatorname{Hom}_G(U, V)$):

$$\dim \mathbf{FinVect}^G(U, V) = \langle \chi_U, \chi_V \rangle.$$

(T7) **Direct sums**: Characters of direct sums are sums of characters; i.e.

$$\chi_{U \oplus V} = \chi_U + \chi_V.$$

(T8) **Equivalence of irreducible modules**: In the context where U, V are irreducible,

$$U \simeq V \iff \langle \chi_U, \chi_V \rangle = 1$$

and furthermore, in the same context,

$$U \not\simeq V \iff \langle \chi_U, \chi_V \rangle = 0.$$

See (G5) for a more concise statement.

(T9) **Equivalence of modules**:

$$U \simeq V \iff \chi_U = \chi_V.$$

(T10) **Frobenius reciprocity**: See (T2).

(T11) **Kernels are kernels**: The kernel of V's character and its representation agree; that is,

$$\operatorname{Ker} \chi_I = \operatorname{Ker} \rho_I.$$

(In the above, we mean the kernel of the group homomorphism ρ_I, *not* the kernel of the linear map which extends ρ_I to a function $\mathbb{C}[G] \to \mathbf{FinVect}(V, V)$.)

(T12) **Normal subgroups**: For $N \subseteq G$,

$$N \trianglelefteq G \iff \exists \mathcal{J} \subseteq \mathcal{I}(G).\ N = \bigcap_{I \in \mathcal{J}} \operatorname{Ker} \chi_I.$$

(T13) **Irreducible factors**: For I irreducible,

$$\text{a factor of } I \text{ appears in } V \iff \langle \chi_I, \chi_V \rangle \neq 0.$$

Furthermore, this expression counts the multiplicity of I in V:

$$V \simeq \bigoplus_{I \in \mathcal{I}(G)} I^{\oplus m_I} \quad \rightsquigarrow \quad \langle \chi_I, \chi_V \rangle = m_j$$

(A factor "appearing" means that $V \simeq I \oplus (\text{stuff})$.)

(T14) **Irreducibility**: See also (G5).

$$V \text{ is irreducible} \iff \langle \chi_V, \chi_V \rangle = 1.$$

(T15) **Order of centralisers**: For $g \in G$,

$$|\mathcal{CN}(g)| = \sum_{I \in \mathcal{I}(G)} |\chi_I(g)|^2.$$

(T16) **Order of group**:

$$|G| = \sum_{I \in \mathcal{I}(G)} |\chi_I(1_G)|^2 = \sum_{I \in \mathcal{I}(G)} (\dim I)^2 .$$

2 The Story of Characters, Told in Broad Strokes

If you want, this section charts the "etymology" of each word in the dictionary!

The slogan is that **characters decategorify modules**, in analogy to how **dimension decategorifies vector spaces**.

$$\begin{array}{ccc} \text{Categorified} & \xrightarrow{\text{character}} & \text{Decategorified} \\ & & \text{What is lost?} \\ \\ \text{Module } U & \longmapsto & \text{List of numbers } \chi_U \in \mathbb{C}^G \\ \Big\downarrow{\scriptstyle \text{morphism space}} & & \Big\downarrow{\scriptstyle \text{its dimension}} \\ \text{Module } V & \longmapsto & \text{List of numbers } \chi_V \in \mathbb{C}^G \end{array} \qquad \begin{array}{c} \text{Algebraic data (action } G \to \mathrm{GL}(U)) \\ \Big\downarrow{\scriptstyle \text{transformation of maps}} \\ \text{Algebraic data: (action } G \to \mathrm{GL}(V)) \end{array}$$

To exemplify the analogy:

$$\begin{array}{ccc} \text{Categorified} & \xrightarrow{\text{dimension}} & \text{Decategorified} \\ & & \text{What is lost?} \\ \\ \text{Vect. Sp. } U & \longmapsto & \text{Number } \dim U \in \mathbb{Z}_{\geq 0} \\ \Big\downarrow{\scriptstyle \text{morphism space}} & & \Big\downarrow{\scriptstyle \text{rank-nullity statements}} \\ \text{Vect. Sp. } V & \longmapsto & \text{Number } \dim V \in \mathbb{Z}_{\geq 0} \end{array} \qquad \begin{array}{c} \text{Algebraic data (vectors)} \\ \Big\downarrow{\scriptstyle \text{transformation of algebraic data}} \\ \text{Algebraic data (vectors)} \end{array}$$

In both situations, **the decategorification classifies the categorification up to isomorphism** – that is, two (finite-dimensional) vector spaces are isomorphic just when they have the same dimension, and two (finite-dimensional, complex) G-modules are isomorphic just when they have the same character. At the level of objects, the decategorification retains all necessary information up to isomorphism (as one would hope of anything being granted the monacre of "decategorification"). However, passing to this "more computational invariant" has the deficiency of losing algebraic data. On decategorifying **FinVect**, we lose information on how linear maps put together vector spaces. On decategorifying $\mathbf{FinVect}^G$, we lose understanding of the actual actions of G and how these actions intertwine.

2.1 Prologue: The Birth of Characters

We now tell a fiction on the story of character theory, all in broad strokes. Long ago, vector spaces were decategorified by their dimension and lived together in harmony. Then, everything changed when algebraists constructed group representations. Only decategorification could tame these, but it vanished. A hundred years passed[2] and all that was known were some parallels:

	Vector spaces	Modules
can be broken up into	one-dimensional subspaces,	irreducible submodules;
the maps between which are	effectively scalars,	effectively scalars;
and which form all objects by	direct sum,	direct sum.

The algebraists believed that they could save the world, and attempted to decategorify the category of G-modules $\mathbf{FinVect}^G$ by mirroring the dimension decategorification of **FinVect**. Fortunately, in the context of a group representation, the dimension appears naturally:

$$G \xrightarrow{\rho_V} \mathrm{GL}(V) \quad \rightsquigarrow \quad \rho_V(1_G) = 1_V = \underbrace{\begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix}}_{n \times n} \quad \rightsquigarrow \quad \dim(V) = n = \mathrm{tr}(\rho_V(1_G)).$$

[2] Not at all. This is a fiction.

This is a natural choice because the representing linear transformations $\rho_V(g), g \in G$ are the only extra data present within objects of $\mathbf{FinVect}^G$, and the only such transformation guaranteed to exist is $\rho_V(1_G) = 1_V$. The generalisation easily reveals itself – the **character** of V is the function $\chi_V : G \to \mathbb{C}$ given by $\chi_V := \operatorname{tr} \circ \rho_V$. The miracle is that this successfully decategorifies away enough complexity to make tractable progress in understanding the category $\mathbf{FinVect}^G$.

Let us demystify this miracle somewhat. For irreducible $I : \mathbf{FinVect}^G$, Schur's lemma provides us with an isomorphism

$$\mathbb{C} \simeq_{\mathbf{Vect}} \mathbf{FinVect}^G(I, I), \qquad \lambda \mapsto \lambda \cdot 1_I.$$

However, given $f \in \mathbf{FinVect}^G(I, I)$ and the promise that $f = \lambda \cdot 1_I$, there is a question of how to find λ. An obvious answer is to evaluate $f(x) = \lambda x$ for a nonzero vector $x \in I$, but this makes for an unnatural choice. A more symmetric option, though it requires accounting for $\dim I$, is to take the trace:

$$\mathbf{FinVect}^G(I, I) \simeq_{\mathbf{Vect}} \mathbb{C}, \qquad f = \lambda \cdot 1_I \mapsto \operatorname{tr}(f) = \lambda \cdot \dim I.$$

That is to say, the trace completely characterises endomorphisms of irreducible modules. Schur's lemma also tells us that the homomorphisms between inequivalent irreducible modules are all zero, so it is fair to say that the trace completely characterises all morphisms between irreducible modules. We now turn to painting this connection more clearly.

2.2 Chapter One: Separating Modules with Inner Products

The preceding text is best understood as a prologue to our story – "Chapter 0", if you will. In chapter one, we begin to understand the hom-spaces $\mathbf{FinVect}^G(U, V)$ for $U, V : \mathbf{FinVect}^G$. Every part of our story takes place over the category of vector spaces, so it is not a surprise to find that $\mathbf{FinVect}^G(U, V)$ is a vector space. Recording its dimension would be a first feat. In perhaps a stroke of luck, or perhaps by arcane insight dug up from historical texts[3], one can find some inspiration in a superficial[4] look at $\mathbf{FinVect}^G(-, -)$:

$$\begin{array}{lll} \text{Categorified} & \mathbf{FinVect}^G \times \mathbf{FinVect}^G \longrightarrow \mathbf{FinVect} & \\ & (U, V) \longmapsto \mathbf{FinVect}^G(U, V), & \\ \text{character} \downarrow & & \\ \text{Decategorified} & \mathbb{C}^G \times \mathbb{C}^G \dashrightarrow \mathbb{C} & \\ & (\chi_U, \chi_V) \longmapsto ?. & \end{array}$$

The superficial symmetry is that $\mathbf{FinVect}^G(-, -)$ appears to decategorify into something resembling an inner product. There is a natural choice of inner product on the set of functions $G \to \mathbb{C}$, given by

$$\begin{aligned} \mathbb{C}^G \times \mathbb{C}^G &\xrightarrow{\langle -, - \rangle} \mathbb{C} \\ (\phi, \psi) &\longmapsto \frac{1}{|G|} \sum_{g \in G} \phi(g) \cdot \overline{\psi(g)}. \end{aligned}$$

The division by $|G|$ may be seen as a normalisation factor; this had proven extremely useful earlier. The miracle of chapter one is that *this inner product idea works* and gives rise to a rich theory, eventually resulting in the dimension formula

$$\dim \mathbf{FinVect}^G(U, V) = \langle \chi_U, \chi_V \rangle$$

and a mechanism to approach other questions in representation theory. How these results follow is not immediate, and will take us another chapter to describe.

[3] The *Module-Wrestler Online Character Dictionary* keeps no good record of history.

[4] Part of the superficiality is not adequately keeping track of the opposite category.

2.3 Chapter Two: Orthonormality of Irreducible Characters

A first observation is that

Lemma 2.1 (Characters of direct sums decompose as sums of characters)
Let $\{V_\alpha\}_{\alpha\in\mathcal{A}}$ be objects of $\mathbf{FinVect}^G$. Then, $\chi_{\oplus_{\alpha\in\mathcal{A}}V_\alpha} = \sum_{\alpha\in\mathcal{A}}\chi_{V_\alpha}$.

Since any module decomposes uniquely into a direct sum of irreducible submodules, we should first study the characters χ_I where I is irreducible. If our inner product $\langle -,-\rangle$ really is to count dimensions of hom-spaces, then Schur's lemma tells us we should expect that

$$\begin{aligned} U \simeq V \text{ are irreducible } &\implies \mathbf{FinVect}^G(U,V) \simeq_{\mathbf{FinVect}} \mathbb{C}^1 \implies \langle\chi_U,\chi_V\rangle = 1, \\ U \not\simeq V \text{ are irreducible } &\implies \mathbf{FinVect}^G(U,V) \simeq_{\mathbf{FinVect}} 0 \implies \langle\chi_U,\chi_V\rangle = 0, \end{aligned}$$

which is to say in other words that the irreducible characters $\{\chi_I \mid I \in \mathcal{I}(G)\}$ are orthonormal. To explain the notation, $\mathcal{I}(G)$ is a set recording one irreducible G-module from each isomorphism class (ignore size issues; they can be fixed). Sure enough,

Proposition 2.2 (Orthonormality of irreducible characters)
For irreducible modules $I, J : \mathbf{FinVect}^G$,

$$\langle\chi_I,\chi_J\rangle = \mathbf{1}_{I\simeq J},$$

with the $\mathbf{1}$ denoting an indicator function. (Perhaps think of the Kronecker delta.)

Now, write two direct sum decompositions $U \simeq \bigoplus_{I\in\mathcal{I}(G)} I^{\oplus u_I}$ and $V \simeq \bigoplus_{I\in\mathcal{I}(G)} I^{\oplus v_I}$. Considering the universal properties of the direct sum, Schur's lemma tells us that

$$\begin{aligned} \mathbf{FinVect}^G(U,V) &\simeq \bigoplus_{I,J\in\mathcal{I}(G)} \mathbf{FinVect}^G(I,J)^{\oplus u_I v_J} \\ &\simeq \bigoplus_{I,J\in\mathcal{I}(G)} \left(\mathbb{C}^{\mathbf{1}_{I=J}}\right)^{\oplus u_I v_J} \\ \mathbf{FinVect}^G(U,V) &\simeq \bigoplus_{I\in\mathcal{I}(G)} \mathbb{C}^{\oplus u_I v_I} \\ \implies \quad \dim \mathbf{FinVect}^G(U,V) &= \sum_{I\in\mathcal{I}(G)} u_I v_I \end{aligned}$$

while if we compute the characters, the previous proposition shows us that

$$\begin{aligned} \langle\chi_U,\chi_V\rangle &= \sum_{I,J\in\mathcal{I}(G)} u_I v_J \langle\chi_I,\chi_J\rangle \\ &= \sum_{I,J\in\mathcal{I}(G)} u_I v_J \mathbf{1}_{I=J} \\ \langle\chi_U,\chi_V\rangle &= \sum_{I\in\mathcal{I}(G)} u_I v_I \end{aligned}$$

and therefore,

Theorem 2.3 (Inner products of characters decategorify hom-spaces)
For $U, V : \mathbf{FinVect}^G$, we have

$$\langle\chi_U,\chi_V\rangle = \dim \mathbf{FinVect}^G(U,V).$$

What we have seen also gives us a strategy to extract from a character $\chi_V = \sum_{I\in\mathcal{I}(G)} v_I \cdot \chi_I$ the coefficients $v_I \in \mathbb{Z}_{\geq 0}$ recording the "multiplicities" in the direct sum $V \simeq \bigoplus_{I\in\mathcal{I}(G)} I^{\oplus v_I}$ of irreducible modules. One can either leverage orthonormality of the irreducible characters (to extract the coefficients straight away), or leverage the fact that $v_I = \dim \mathbf{FinVect}^G(I,V)$ (which is not hard to prove). Either way,

Theorem 2.4 (Irreducible characters give an orthonormal basis for all characters)
Just as the irreducible submodules give a "direct sum basis" for the finite-dimensional complex G-modules up to isomorphism, the irreducible characters give a "$\mathbb{Z}_{\geq 0}$-weighted basis" for their characters. That is, we have a correspondence

$$\mathbf{FinVect}^G_{/\simeq} \longleftrightarrow \{\text{characters of } G\}$$
$$V \simeq \bigoplus_{I \in \mathcal{I}(G)} I^{\oplus a_I} \longleftrightarrow \chi_V = \sum_{I \in \mathcal{I}(G)} a_I \cdot \chi_I.$$

Read the right way, the previous result entails

Theorem 2.5 (Equality of characters decategorifies equivalence of modules)
For $U, V : \mathbf{FinVect}^G$, we have

$$U \simeq V \iff \chi_U = \chi_V.$$

Our result 2.7 on a "basis" for the set of characters has a linear algebraic deficiency. It would be nice to interpret this as an actual basis for a vector space, but the set of characters is not closed under scaling by $\mathbb{C}$. Rather than doing something artificial (e.g. taking the $\mathbb{C}$-span of this set), we can consider the vector space $\mathbb{C}^G$. Thinking categorically, we find that

Lemma 2.6 (Characters are class functions)
Just as conjugate (isomorphic) elements x and gxg^{-1} of a group have similar (isomorphic) representing matrices $\rho_V(x)$ and $\rho_V(g)\rho_V(x)\rho_V(g)^{-1}$, the character assigns these elements equal values. That is, $\chi_V(x) = \chi_V(gxg^{-1})$. (This decategorifies isomorphism into equality.)

This does mean that characters cannot span $\mathbb{C}^G$ for non-Abelian G. However, it also shows us a better target for their span.

Theorem 2.7 (Irreducible characters give an orthonormal basis for class functions)
The set $\{\chi_I \mid I \in \mathcal{I}(G)\}$ is an orthonormal basis for $\mathbb{C}^{\mathcal{CC}(G)}$. There is hence one irreducible character of G for each conjugacy class of G. Categorifying, there is (up to isomorphism) one irreducible module of G for each conjugacy class of G.

We can now conclude that $\mathcal{I}(G)$ is a finite set.

2.4 Chapter Three: Bonus Goodies

All good things come in threes, and the missing partner to characters and conjugacy is the left-regular representation $\mathbb{C}[G]$. A telling of this story which does not take such broad strokes would actually reveal, in showing that the irreducible characters span $\mathbb{C}^{\mathcal{CC}(G)}$ (as part of proving 2.7), that the left-regular representation appears. To take yet another broad stroke, the conclusion here is that this representation can be completely torn apart, and it summarises useful information about irreducible representations.

Proposition 2.8 (The left-regular representation summarises the irreducible modules)
The left-regular representation $\mathbb{C}[G]$ decomposes as

$$\mathbb{C}[G] \simeq \bigoplus_{I \in \mathcal{I}(G)} I^{\oplus \dim I},$$

or in the language of characters,

$$\chi_{\mathbb{C}[G]} = \sum_{I \in \mathcal{I}(G)} \dim I \cdot \chi_I.$$

Furthermore, the centre of the endomorphism algebra captures the conjugacy classes and irre-

ducible modules together:

$$|\mathcal{CC}(G)| = \dim\left(Z\left(\mathbf{FinVect}^G(\mathbb{C}[G],\mathbb{C}[G])\right)\right) = |\mathcal{I}(G)|.$$

This gives another proof, independent of 2.7, that irreducible characters biject with conjugacy classes.

If your goal is to apply representation theory to distinguish groups, the story told thus far should guide you towards finding characters. The "holy grail" in this avenue is to list the character table of any finite groups you care about (see section 3), and point at different tables to tell apart many groups. However, before we leave you to such calculations, let us close this story with some words on induction and restriction.

Theorem 2.9 (Frobenius reciprocity)
Fix $H \leq G$ in **FinGroup** and an enumeration $\{x_\alpha H\}_{\alpha\in\mathcal{A}}$ of the cosets in G/H. The adjoint functors $\mathrm{Ind}_H^G \dashv \mathrm{Res}_H^G$ act in characters by

$$\begin{array}{ccccc} \sum_{\alpha\in\mathcal{A}} \hat{\chi}_U(x_\alpha^{-1}(-)x_\alpha) & & \{\text{char's of } G\} & \xleftarrow{\text{char.}} \mathbf{FinVect}^G & \xrightarrow{\text{char.}} \{\text{char's of } G\} \\ \uparrow & & \uparrow & \mathrm{Ind}_H^G \uparrow \;\big|\; \mathrm{Res}_H^G \downarrow & \downarrow \\ \chi_U & & \{\text{char's of } H\} & \xleftarrow[\text{char.}]{} \mathbf{FinVect}^H & \xrightarrow[\text{char.}]{} \{\text{char's of } H\} \end{array} \qquad \begin{array}{c} \chi_V \\ \downarrow \\ \chi_V|_H \end{array}$$

where $\hat{\chi}_U : G \to \mathbb{C}$ is given by $\hat{\chi}_U(h) = \chi_U(h)$ for $h \in H$ and $\hat{\chi}_U(g) = 0$ for $g \in G \setminus H$. This gives a formula to define induction and restriction operators on the space of class functions,

$$\begin{array}{ccc} \sum_{\alpha\in\mathcal{A}} \hat{\varphi}(x_\alpha^{-1}(-)x_\alpha) & & \mathbb{C}^{\mathcal{CC}(G)} \\ \uparrow & & \mathrm{ind}_H^G \uparrow \;\big|\; \mathrm{res}_H^G \downarrow \\ \varphi & & \mathbb{C}^{\mathcal{CC}(H)} \end{array} \qquad \begin{array}{c} \psi \\ \downarrow \\ \psi|_H \end{array},$$

and the formula for ind_H^G is independent of the representatives x_α. The fact that the functors are adjoint appears in characters as the linear operators being Hermitian adjoints.

$$\mathbf{FinVect}^G\left(\mathrm{Ind}_H^G U, V\right) \simeq \mathbf{FinVect}^H\left(U, \mathrm{Res}_H^G V\right) \quad \rightsquigarrow \quad \langle \mathrm{ind}_H^G \chi_U, \chi_V\rangle_G = \langle \chi_U, \mathrm{res}_H^G \chi_V\rangle_H$$

3 Some Character Tables

3.1 Baby examples

Trivial group $S_1 \simeq \mathbb{Z}_{/1}$:

S_1	()
χ_{trivial}	1

Order two group $S_2 \simeq \mathbb{Z}_{/2}$:

S_2	()	$(**)$
χ_{trivial}	1	1
χ_{sign}	1	-1

Order three group $\mathbb{Z}_{/3}$:

$\mathbb{Z}_{/3}$	0	1	2
χ_0	1	1	1
χ_1	1	$e^{2\pi i/3}$	$e^{-2\pi i/3}$
χ_2	1	$e^{-2\pi i/3}$	$e^{2\pi i/3}$

Order four groups:

$\mathbb{Z}_{/4}$	0	1	2	3
χ_0	1	1	1	1
χ_1	1	i	-1	$-i$
χ_2	1	-1	1	-1
χ_3	1	$-i$	-1	i

$\mathbb{Z}_{/2} \times \mathbb{Z}_{/2}$	00	01	10	11
χ_{00}	1	1	1	1
χ_{01}	1	-1	1	-1
χ_{10}	1	1	-1	-1
χ_{11}	1	-1	-1	1

Order five group $\mathbb{Z}_{/5}$; here, ω is a primitive fifth root of unity:

$\mathbb{Z}_{/5}$	0	1	2	3	4
χ_0	1	1	1	1	1
χ_1	1	ω	ω^2	ω^3	ω^4
χ_2	1	ω^2	ω^4	ω	ω^3
χ_3	1	ω^3	ω	ω^4	ω^2
χ_4	1	ω^4	ω^3	ω^2	ω

Order six groups $\mathbb{Z}_{/6}$ and $S_3 \simeq D_3$; here, ω is a primitive sixth root of unity:

$\mathbb{Z}_{/6}$	0	1	2	3	4	5
χ_0	1	1	1	1	1	1
χ_1	1	ω	ω^2	ω^3	ω^4	ω^5
χ_2	1	ω^2	ω^4	1	ω^2	ω^4
χ_3	1	-1	1	-1	1	-1
χ_4	1	ω^4	ω^2	1	ω^4	ω^2
χ_5	1	ω^5	ω^4	ω^3	ω^2	ω

S_3	()	$(**)$	$(***)$
$\chi_{(3)}$	1	1	1
$\chi_{(2,1)}$	2	0	-1
$\chi_{(1,1,1)}$	1	-1	1

Order seven group $\mathbb{Z}_{/7}$; here, ω is a primitive seventh root of unity:

$\mathbb{Z}_{/7}$	0	1	2	3	4	5	6
χ_0	1	1	1	1	1	1	1
χ_1	1	ω	ω^2	ω^3	ω^4	ω^5	ω^6
χ_2	1	ω^2	ω^4	ω^6	ω^1	ω^3	ω^5
χ_3	1	ω^3	ω^6	ω^2	ω^5	ω^1	ω^4
χ_4	1	ω^4	ω^1	ω^5	ω^2	ω^6	ω^3
χ_5	1	ω^5	ω^3	ω^1	ω^6	ω^4	ω^2
χ_6	1	ω^6	ω^5	ω^4	ω^3	ω^2	ω^1

3.2 Quaternion group

FIXME: I have no idea how to start this one!

3.3 Abelian groups

The character tables of the cyclic groups are easy to describe. Fix $n \in \mathbb{Z}_{>0}$ and let ω be a primitive nth root of unity. There is one irreducible character χ_x for each element $x \in \mathbb{Z}_{/n}$, and its values are $\chi_x(y) = \omega^{xy}$. So, the character table is

$\mathbb{Z}_{/n}$	0	1	2	$\cdots$	$n-1$
χ_0	1	1	1	$\cdots$	1
χ_1	1	ω	ω^2	$\cdots$	ω^{n-1}
χ_2	1	ω^2	ω^4	$\cdots$	ω^{n-2}
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$
χ_{n-1}	1	ω^{n-1}	ω^{n-2}	$\cdots$	ω^1

The irreducible representations are also easy to describe. They are exactly the characters above; think of the scalars as scaling the identity map on $\mathbb{C}$.

Consider now a product $\mathbb{Z}_{/n_1} \times \cdots \times \mathbb{Z}_{/n_k}$ of cyclic groups. Let ω_j be a primitive n_jth root of unity for each $j \in \{1, \ldots, k\}$. There is one irreducible character for each element $(x_1, \ldots, x_k) \in \mathbb{Z}_{/n_1} \times \cdots \times \mathbb{Z}_{/n_k}$, and it is given by

$$\chi_{x_1,\ldots,x_k}(y_1, \ldots, y_k) = \omega_1^{x_1 y_1} \cdots \omega_k^{x_k y_k}.$$

Again, the irreducible representations themselves are exactly the irreducible characters.

3.4 Dihedral groups

The dihedral group D_n is the graph automorphism group of the n-cycle. The graph's eigenspaces provide many irreducible representations of D_n. If we present $D_n = \langle r, s \mid r^n = s^2 = 1 \text{ and } srs = r^{-1} \rangle$, then the eigenspaces provide every irreducible representation except for the "partners" of the one-dimensional representations. If n is odd, then an eigenspace provides the trivial representation; multiplying by the one-dimensional "sign of s" representation $r \mapsto 1, s \mapsto -1$ gives the missing partner. If n is even, then we still need this partner to the trivial representation, but we also need a partner to the "half-way around / sign of r" representation $r \mapsto -1, s \mapsto 1$; this partner is obtained in the same way, and it is $r \mapsto -1, s \mapsto -1$. FIXME: I'm currently way too lazy to do it, but I really should include some pictures of these irreps and give a few ($n \in \{3, 4, 5, 6\}$) character tables explicitly; they're quite beautiful.

3.5 Symmetric groups

The irreducible representations of the symmetric groups are the Specht modules. FIXME: I should include a few small character tables and a note about the calculation methods; way too lazy right now lol (look at the timestamp on this git commit).

3.6 Alternating groups

FIXME: I actually have no idea where to start with these

4 Proofs

All proofs are collected here. They are all reproduced by myself for my own practice. We assume results from representation theory which occur "prior to character theory" – for example, Schur's lemma and Maschke's theorem. Throughout, $G : \mathbf{FinGroup}$ and $U, V : \mathbf{FinVect}^G$ unless otherwise specified.

4.1 Proof that characters are class functions

$\chi_V(gxg^{-1}) = \operatorname{tr}(\rho_V(g)\rho_V(x)\rho_V(g)^{-1}) = \operatorname{tr}(\rho_V(x)) = \chi_V(x)$ for any $x, g \in G$. ∎

4.2 Proof that characters of inverses are conjugates of characters (G3)

Fix $g \in G$. We wish to show that $\chi_V(g^{-1}) = \overline{\chi_V(g)}$. Characters are traces, so $\chi_V(g^{-1})$ is the sum of eigenvalues of $\rho(g^{-1}) = \rho(g)^{-1}$, and thus is the sum $\sum_\lambda 1/\lambda$ for eigenvalues λ of $\rho(g)$ (counted with multiplicity). So, we will be done if we can show that each such λ lies on the unit circle, for then $1/\lambda = \overline{\lambda}$. Indeed, if we let n be the (finite!) order of g, then $g^n = 1_G$ implies that $\lambda^n = 1$. Being a root of unity, λ lies on the unit circle. ∎

4.3 Proof that irreducible characters are orthonormal: (G5), (T8)

Fix $I, J \in \mathcal{I}(G)$. By Schur's lemma, the task is essentially to show $\langle \chi_I, \chi_J \rangle = \dim \mathbf{FinVect}^G(I, J)$. To make our lives easier, identify $I \simeq \mathbb{C}^m$ and $J \simeq \mathbb{C}^n$ with matrix representations $M : G \to \mathrm{GL}\,\mathbb{C}^m$ and $N : G \to \mathrm{GL}\,\mathbb{C}^n$. We will denote the entries of a matrix f by f_i^j, with i indexing the columns and j indexing the rows; in particular, $M(g)_i^j \in \mathbb{C}$ denotes an entry of $M(g)$.

Notice first that the "averaging trick" turns vector space morphisms to module morphisms:

$$f \in \mathbf{FinVect}(I, J) \implies \frac{1}{|G|} \sum_{g \in G} \rho_J(g) \circ f \circ \rho_I(g^{-1}) \in \mathbf{FinVect}^G(I, J)$$

and the requisite commutative diagram is easily checked. A cheeky way to encode this by matrix representation is

$$f \in \mathrm{Mat}_{n \times m}(\mathbb{C}) \implies \frac{1}{|G|} \sum_{g \in G} N(g) \circ f \circ M(g^{-1}) \text{ is a module homomorphism.}$$

Doing things this way, we can play a bit more of a trick and consider "all of these at once" by introducing formal variables. Let $\left\{ a_i^j \;\middle|\; i \in \{1, \ldots, m\} \text{ and } j \in \{1, \ldots, n\} \right\}$ be a set of distinct formal variables, and as the notation suggests, consider the matrix $a \in \mathrm{Mat}_{n \times m}\,\mathbb{C}[\text{all } a_i^j\text{'s}]$ (the indices i and j are understood to run over all $i \in \{1, \ldots, m\}$ and $j \in \{1, \ldots, n\}$ in the preceding notation; sorry, it's horrible). We denote the "averaged version" of this "formal map" by α; more precisely,

$$\alpha_i^j := \frac{1}{|G|} \sum_{g \in G} N(g)_v^j a_u^v M(g^{-1})_i^u \in \mathbb{C}[\text{all } a_i^j\text{'s}], \qquad \alpha := \left(\alpha_i^j\right)_{i,j} \in \mathrm{Mat}_{n \times m}\,\mathbb{C}[\text{all } a_i^j\text{'s}],$$

with implicit summation on repeated indices (we could also use this convention to sum over the repeated g, but elect not to). Then, assignment of values f_i^j to variables a_i^j defines actual module morphisms $\alpha|_{a_i^j := f_i^j} \in \mathbf{FinVect}^G(I, J)$. The formal matrix α is the zero matrix of $\mathrm{Mat}_{n \times m}\,\mathbb{C}[\text{all } a_i^j\text{'s}]$ just when the matrices $\alpha|_{a_i^j := f_i^j}$ are all zero the matrix in $\mathrm{Mat}_{n \times m}\,\mathbb{C}$ – i.e. all the zero map in $\mathbf{FinVect}^G(I, J)$. The rest of this proof is an algebraic examination of what happens to α upon applying Schur's lemma.

In the case where $I \not\simeq J$...

Each $\alpha|_{a_i^j := f_i^j}$ is the zero map, and we conclude that $\alpha = 0$. Now, for all $i, u \in \{1, \ldots, m\}$ and $j, v \in \{1, \ldots, n\}$, if we read the coefficient of a_v^u in α_i^j, we find

$$\begin{aligned} 0 = \alpha_i^j &= \frac{1}{|G|} \sum_{g \in G} N(g)_v^j a_u^v M(g^{-1})_i^u \\ \implies \quad 0 &= \frac{1}{|G|} \sum_{g \in G} N(g)_v^j M(g^{-1})_i^u \qquad \forall i, j, u, v. \end{aligned}$$

We can now put $u := i$ and $v := j$, and sum on i, j to complete this branch of the proof.

$$\begin{aligned} \implies \quad 0 &= \frac{1}{|G|} \sum_{g \in G} \sum_j N(g)_j^j \cdot \sum_i M(g^{-1})_i^i \\ &= \frac{1}{|G|} \sum_{g \in G} \operatorname{tr}(\rho_J(g)) \cdot \operatorname{tr}(\rho_I(g^{-1})) \\ &= \frac{1}{|G|} \sum_{g \in G} \operatorname{tr}(\rho_J(g)) \cdot \operatorname{tr}(\rho_I(g^{-1})) \\ &= \frac{1}{|G|} \sum_{g \in G} \chi_J(g) \cdot \chi_I(g^{-1}) \\ &= \frac{1}{|G|} \sum_{g \in G} \chi_J(g) \cdot \overline{\chi_I(g)} \\ 0 &= \langle \chi_J, \chi_I \rangle \\ \implies \quad 0 &= \langle \chi_I, \chi_J \rangle . \end{aligned}$$

In the case where $I \simeq J \ldots$

Simplifying our lives again, assume without loss of generality that $I = J$ (so $m = n$ and $M = N$). Schur's lemma tells us that each $\alpha|_{a_i^j := f_i^j} = \lambda_f \operatorname{id}_I$ for scalar multiples λ_f. Denoting by f the linear transformation on $I = \mathbb{C}^m$ given by the matrix $(f_i^j)_{i,j}$, the scalar λ_f itself may be recovered by

$$\begin{aligned} \lambda_f &= \frac{1}{m} \operatorname{tr}(\lambda_{f_i^j} \operatorname{id}_I) \\ &= \frac{1}{m} \operatorname{tr}\left(\alpha|_{a_i^j := f_i^j} \right) \\ &= \frac{1}{m|G|} \sum_{g \in G} \operatorname{tr}\left(\rho_I(g) \circ f \circ \rho_I(g^{-1}) \right) \\ &= \frac{1}{m|G|} \sum_{g \in G} \operatorname{tr} f \\ \lambda_f &= \frac{1}{m} f_i^i. \end{aligned}$$

We can use this formula to express α as follows.

$$\alpha|_{a_i^j := f_i^j} = \lambda_f \operatorname{id}_I = \frac{1}{m} f_i^i \operatorname{id}_I \quad \implies \quad \alpha = \frac{1}{m} a_i^i \operatorname{id}_{m \times m}$$

The rest of this argument is chasing down coefficients of the a_u^v. Unless summed over, the variables i, j, u, v are arbitrary elements of $\{1, \ldots, n\}$.

$$\frac{1}{m} \delta_u^v \delta_i^j = \frac{1}{|G|} \sum_{g \in G} M(g)_v^j M(g^{-1})_i^u$$

$$\implies \quad \frac{1}{m}\delta_i^j\delta_i^j = \frac{1}{|G|}\sum_{g\in G} M(g)_j^j M(g^{-1})_i^i$$
$$\implies \quad 1 = \frac{1}{|G|}\sum_{g\in G} \operatorname{tr}(\rho_I(g))\cdot\operatorname{tr}(\rho_I(g^{-1}))$$
$$\implies \quad 1 = \langle \chi_I, \chi_I\rangle$$

Having dealt with both cases, we are finished with this proof. ∎

4.4 Proof that irreducible characters give an orthonormal basis for class functions: (G6)

The irreducible characters are orthonormal and hence linearly independent. As the characters are class functions, it remains only to show that the set $\beta := \{\chi_I \mid I \in \mathcal{I}(G)\}$ of irreducible characters spans $\mathbb{C}^{\mathcal{C}\mathcal{C}(G)}$. If[5] we put $L := \operatorname{span}\beta$, then $\mathbb{C}^{\mathcal{C}\mathcal{C}(G)} = L \oplus L^\perp$ and so it is enough to show that $L^\perp = 0$.

Fix $f \in L^\perp$. Schur's lemma tells us that for $I \in \mathcal{I}(G)$, we have an isomorphism

$$\mathbb{C} \simeq_{\mathbf{FinVect}} \mathbf{FinVect}^G(I, I)$$
$$\lambda \mapsto \lambda \cdot 1_I$$

and so, because $\dim I \neq 0$, we get that the trace is also an isomorphism

$$\operatorname{tr} : \mathbf{FinVect}^G(I, I) \simeq_{\mathbf{FinVect}} \mathbb{C}.$$

The rest of the proof now falls into place because the characters are given in terms of traces, and the isomorphism lets us "undo" the trace operation. To be more precise, observe that

$$0 = \langle f, \chi_I\rangle = \frac{1}{|G|}\sum_{g\in G} f(g)\operatorname{tr}(\rho_I(g^{-1})) = \operatorname{tr}\left(\frac{1}{|G|}\sum_{g\in G} f(g)\rho_I(g^{-1})\right)$$
$$\implies \quad 0 = \frac{1}{|G|}\sum_{g\in G} f(g)\rho_I(g^{-1}).$$

In the above, we must technically check that $\frac{1}{|G|}\sum_{g\in G} f(g)\rho_I(g^{-1})$ is actually G-equivariant; this is not hard. Now, for $V : \mathbf{FinVect}^G$, we can decompose V into irreducible submodules and apply the above to find that

$$\frac{1}{|G|}\sum_{g\in G} f(g)\rho_V(g^{-1}) = 0.$$

Setting $V := \mathbb{C}[G]$ to be the left-regular representation, the transformations $\rho_V(g^{-1})$ are now linearly independent, so the coefficients $f(g)$ must all be zero. That is, $f = 0$ and hence $L^\perp = 0$. ∎

4.5 Proof that the class number counts irreducible modules: (T3)

Read this from any of 2.7, 2.8, or (G6). ∎

4.6 Proof that number of irreducible modules count whether a group is Abelian: (T1)

G is Abelian just when each conjugacy class has size one, and hence just when there are $|G|$ conjugacy classes. This number is also counted by $|\mathcal{I}(G)|$. ∎

[5] This argument is a re-framing of that at on groupprops.

4.7 Proof of character table column orthogonality and orders of centralisers: (G1), (T15)

The goal is essentially to understand the various dot products $\sum_{I \in \mathcal{I}(G)} \chi_I(g)\overline{\chi_I(h)}$ for $g, h \in G$. Fixing h and letting g vary, we have a class function $f := \sum_{I \in \mathcal{I}(G)} \overline{\chi_I(h)}\chi_I$. The goal is essentially to show that $f = F$, for $F := |\mathcal{CN}(h)| \cdot 1_{-\in \mathcal{CL}(h)}$. Since the irreducible characters form a basis and f is already written as a linear combination of irreducible characters, it is enough to calculate the coefficients F_I in the expansion $F =: \sum_{I \in \mathcal{I}(G)} F_I \chi_I$. Fixing $I \in \mathcal{I}(G)$,

$$\begin{aligned} F_I &= \langle F, \chi_I \rangle \\ &= \frac{1}{|G|} \sum_{g \in G} |\mathcal{CN}(h)| \cdot 1_{g \in \mathcal{CL}(h)} \cdot \overline{\chi_I(g)} \\ &= \frac{|\mathcal{CN}(h)| \cdot |\mathcal{CL}(h)|}{|G|} \cdot \overline{\chi_I(h)} \\ F_I &= \overline{\chi_I(h)} \end{aligned}$$

and therefore, $F = f$. ∎

4.8 Proof that dimensions of modules are computed by characters: (T5)

Obvious. ∎

4.9 Proof that order of group is square sum of irreducible dimensions: (T16)

Apply (T15) to the element $1_G \in G$, whose centraliser is G, and then use the fact that $\dim(V) = \chi_V(1_G)$. ∎

4.10 Proof that characters of direct sums are sums of characters: (T7)

Obvious. ∎

4.11 Proof that isomorphism of modules is computed by equality of characters: (T9)

Factorise $U \simeq \bigoplus_{I \in \mathcal{I}(G)} I^{\oplus u_I}$ and $V \simeq \bigoplus_{I \in \mathcal{I}(G)} I^{\oplus v_I}$. Then, $\chi_U = \sum_{I \in \mathcal{I}(G)} u_I \cdot \chi_I$ and $\chi_V = \sum_{I \in \mathcal{I}(G)} v_I \cdot \chi_I$. Since $\{\chi_I \mid I \in \mathcal{I}(G)\}$ is linearly independent, we have $\chi_U = \chi_V$ just when $u_I = v_I$ for all I, which itself happens just when $U \simeq V$. ∎

4.12 Proof that dimension of hom space is computed by inner products of characters: (T6)

Factorise $U \simeq \bigoplus_{I \in \mathcal{I}(G)} I^{\oplus u_I}$ and $V \simeq \bigoplus_{I \in \mathcal{I}(G)} I^{\oplus v_I}$. The rest of the proof is solved by applying universal properties of the direct sum, Schur's lemma, and orthonormality of irreducible characters. Specifically, observe that

$$\begin{aligned} \dim \mathbf{FinVect}^G(U, V) &= \sum_{I,J \in \mathcal{I}(G)} u_I \cdot v_J \cdot \dim \mathbf{FinVect}^G(I, J) \\ &= \sum_{I,J \in \mathcal{I}(G)} u_I \cdot v_J \cdot \mathbf{1}_{I \simeq J} \\ &= \sum_{I,J \in \mathcal{I}(G)} u_I \cdot v_J \cdot \langle \chi_I, \chi_J \rangle \\ \dim \mathbf{FinVect}^G(U, V) &= \langle \chi_U, \chi_V \rangle \end{aligned}$$

as desired. ∎

4.13 Proof that irreducibility is computed by inner products of characters: (T14)

Factorise $V \simeq \bigoplus_{I \in \mathcal{I}(G)} I^{\oplus v_I}$. Then, by orthonormality of irreducible characters, $\langle \chi_V, \chi_V \rangle = \sum_{I \in \mathcal{I}(G)} v_I^2$. This is equal to 1 if and only if precisely one v_I is nonzero and takes value $v_I = 1$; this describes exactly when $V \simeq I$ is irreducible. ∎

4.14 Proof that appearance irreducible factors is computed by inner products of characters: (T13)

Factorise $V \simeq \bigoplus_{I \in \mathcal{I}(G)} I^{\oplus v_I}$. Then,

$$\begin{aligned} \langle \chi_I, \chi_V \rangle &= \dim \mathbf{FinVect}^G(I, V) \\ &= \sum_{J \in \mathcal{I}(G)} v_J \cdot \dim \mathbf{FinVect}^G(I, J) \\ \langle \chi_I, \chi_V \rangle &= v_I, \end{aligned}$$

as desired. ∎

4.15 Proof of Frobenius reciprocity: (G4), (T2), (T10)

Fix $H \leq G$. The linear adjunction of ind_H^G to res_H^G can be verified at the level of class functions. Take coset representatives enumerating $G/H =: \{x_\alpha H\}_{\alpha \in \mathcal{A}}$. For $\varphi \in \mathbb{C}^{\mathrm{cc}(H)}$, we have $\mathrm{ind}_H^G(\varphi)(h) := \sum_{\alpha \in \mathcal{A}} \hat{\varphi}(x_\alpha^{-1} h x_\alpha)$, where $\hat{\varphi}(h) := \varphi(h)$ for $h \in H$ and $\hat{\varphi}(g) := 0$ for $g \in G \setminus H$. Now, for $\varphi \in \mathbb{C}^{\mathrm{cc}(H)}$ and $\psi \in \mathbb{C}^{\mathrm{cc}(G)}$, observe that

$$\begin{aligned} \left\langle \mathrm{ind}_H^G \varphi, \psi \right\rangle_G &= \frac{1}{|G|} \sum_{g \in G} \sum_{\alpha \in \mathcal{A}} \hat{\varphi}(x_\alpha^{-1} g x_\alpha) \cdot \overline{\psi(g)} \\ &= \frac{1}{|G|} \sum_{\alpha \in \mathcal{A}} \sum_{g \in G} \hat{\varphi}(x_\alpha^{-1} g x_\alpha) \cdot \overline{\psi(g)} \\ &= \frac{1}{|G|} \sum_{\alpha \in \mathcal{A}} \sum_{g' \in G} \hat{\varphi}(g') \cdot \overline{\psi(x_\alpha g' x_\alpha^{-1})} \\ &= \frac{1}{|G|} \sum_{\alpha \in \mathcal{A}} \sum_{g \in G} \hat{\varphi}(g) \cdot \overline{\psi(g)} \\ &= \frac{|G/H|}{|G|} \sum_{g \in G} \hat{\varphi}(g) \cdot \overline{\psi(g)} \\ &= \frac{1}{|H|} \sum_{g \in G} \hat{\varphi}(g) \cdot \overline{\psi(g)} \\ &= \frac{1}{|H|} \sum_{h \in H} \varphi(h) \cdot \overline{\psi(h)} \\ \left\langle \mathrm{ind}_H^G \varphi, \psi \right\rangle_G &= \langle \varphi, \mathrm{res}_H^G \psi \rangle_H \end{aligned}$$

as desired. Verifying the adjunction of functors $\mathrm{Ind}_H^G \dashv \mathrm{Res}_H^G$ is easy if you take the definition of induction to be $\mathrm{Ind}_H^G := - \otimes_{\mathbb{C}[H]} \mathbb{C}[G]$. We can view the category $\mathbf{FinVect}^G$ of G-modules as isomorphic to the category $\mathbb{C}[G]\text{-}\mathbf{Mod}$ of left $\mathbb{C}[G]$-modules[6]; in the latter category, Ind_H^G may be viewed as an extension of scalars. This provides the required adjunction $\mathrm{Ind}_H^G \dashv \mathrm{Res}_H^G$ when viewing $\mathrm{Ind}_H^G : \mathbb{C}[H]\text{-}\mathbf{Mod} \rightleftarrows \mathbb{C}[G]\text{-}\mathbf{Mod} : \mathrm{Res}_H^G$. We transport this along isomorphisms of categories to find our adjunction $\mathrm{Ind}_H^G \dashv \mathrm{Res}_H^G$ with $\mathrm{Ind}_H^G : \mathbf{FinVect}^H \rightleftarrows \mathbf{FinVect}^G : \mathrm{Res}_H^G$. If you take a different definition of induction (for example, the more computational one related to

[6] $\mathbb{C}[G]$ is a ring, though generically not a commutative one.

coset representatives), then the categorical adjunction is more tricky to verify. **We will not do so here.**

4.16 Proof that kernels are kernels: (T11)

Let n be the order of g. It is easy to show that $\operatorname{Ker}\rho_V \subseteq \operatorname{Ker}\chi_V$. For the reverse inclusion, fix $g \in \operatorname{Ker}\chi_V$, so that $\operatorname{tr}(\rho_V(g)) = \dim V$. The trace itself is a sum of eigenvalues λ of $\rho_V(g)$; as $\rho_V(g)^n = 1_V$, we must have $\lambda^n = 1$ and so $|\lambda| = 1$. By the triangle inequality,

$$\dim V = \left|\sum_\lambda \lambda\right| \geq \sum_\lambda |\lambda| = (\text{num. eigenvalues of } \rho_V(g)) = \dim V.$$

The equality forces each λ to be 1. Consulting a Jordan canonical form for the matrix $\rho_V(g)$ and imposing that $\rho_V(g)^n = 1_V$ on this matrix, we find that $\rho_V(g) = 1_V$. This shows that $g \in \operatorname{Ker}\rho_V$, so $\operatorname{Ker}\chi_V \subseteq \operatorname{Ker}\rho_V$.

4.17 Proof that normal subgroups are intersections of irreducible kernels: (T12)

A normal subgroup $N \trianglelefteq G$ is the kernel of the map $G \to G/N$. Viewing G/N as a subgroup of a permutation group, we get a map $G \to S_n$ with the same kernel, for sufficiently large $n \in \mathbb{Z}_{\geq 0}$. Extending linearly, we get a representation $G \to \operatorname{GL}\mathbb{C}^n$ with equal kernel. Factorise this representation into irreducible submodules $\bigoplus_{I\in\mathcal{I}(G)} I^{\oplus m_I}$. We are now asking for the kernel of a map $G \to \operatorname{GL}\left(\bigoplus_{I\in\mathcal{I}(G)} I^{\oplus m_I}\right)$, which is the same as the intersection of the kernels of maps $\rho_I : G \to \operatorname{GL} I$ for each $I \in \mathcal{I}(G)$ with $m_I \neq 0$. Each $\operatorname{Ker}\rho_I = \operatorname{Ker}\chi_I$ is the kernel of an irreducible character, so we have now written our subgroup as an intersection of kernels of irreducible characters.

4.18 Proof that the commutation subgroup is the intersection of one-dimensional character kernels: (T4)

Let $\mathcal{L} \subseteq \mathcal{I}(G)$ denote the set of one-dimensional irreducible G-modules (rather, one for each isomorphism class). For each $I \in \mathcal{L}$, the endomorphisms of I are multiplication by scalars, so ρ_I has each element of G act by scalars. Thus, for $g, h \in G$, ρ_I carries $ghg^{-1}h^{-1}$ to the scalar 1, and so $[G,G] \subseteq \operatorname{Ker}\rho_I = \operatorname{Ker}\chi_I$. As this held for each $I \in \mathcal{L}$, we have $[G,G] \subseteq \bigcap_{I\in\mathcal{L}} \operatorname{Ker}\chi_I$. It remains to prove the other inclusion.

Our proof in subsection 4.16 gave a recipe to explicitly write any normal subgroup as an intersection of irreducible character kernels; we apply this recipe to the normal subgroup $[G,G]$. After combing through the details, we are after the intersection $\bigcap_{I\in\mathcal{K}} \operatorname{Ker}\chi_I$, where $\mathcal{K} \subseteq \mathcal{I}(G)$ is the set of irreducible submodules appearing in a factorisation of the left coset representation $G \to \operatorname{GL}\mathbb{C}[G/[G,G]]$. If we show that $\mathcal{K} \subseteq \mathcal{L}$, then $[G,G] = \bigcap_{I\in\mathcal{K}} \operatorname{Ker}\chi_I \supseteq \bigcap_{I\in\mathcal{L}} \operatorname{Ker}\chi_I \supseteq [G,G]$ and we will be done.

Consider the set of operators $\mathcal{G} := \{g\cdot - : \mathbb{C}[G/[G,G]] \to \mathbb{C}[G/[G,G]] \mid g \in G\}$. These operators permute the basis $G/[G,G]$ of $\mathbb{C}[G/[G,G]]$, so they are effectively permutation matrices and therefore they diagonalise. Furthermore, these operators commute; this is essentially because all elements commute in $G/[G,G]$, but let us check this more explicitly. Fix $g, h, x \in G$. Then, $h^{-1}g^{-1}hg \in [G,G]$, and normality grants us that $x^{-1}h^{-1}g^{-1}hgx \in [G,G]$ also. Thus,

$$g\cdot h\cdot x[G,G] = ghx[G,G] = ghxx^{-1}h^{-1}g^{-1}hgx[G,G] = hgx[G,G] = h\cdot g\cdot x[G,G].$$

Varying x, we find that $g\cdot h\cdot -$ and $h\cdot g\cdot -$ agree on a basis for $\mathbb{C}[G/[G,G]]$, and therefore are equal functions. Thus, $\mathcal{G}$ is a set of commuting diagonalisable operators, and therefore it simultaneously diagonalises over some basis β for $\mathbb{C}[G/[G,G]]$. Thanks to diagonalisation, G acts by scalars on each element $b \in \beta$; thus, the spaces $\operatorname{span}_{\mathbb{C}}\{b\}$ are each distinct, one-dimensional G-submodules which together span $\mathbb{C}[G/[G,G]]$. This is to say that left coset representation factorises into a direct sum of one-dimensional irreducible G-modules; i.e. $\mathcal{K} \subseteq \mathcal{L}$.

4.19 Proof of how the left-regular representation factorises: 2.8

The character of the left-regular representation is $\chi_{\mathbb{C}[G]} = (g \mapsto |G| \cdot \mathbf{1}_{g=1_G})$. For each $I \in \mathcal{I}(G)$, the multiplicity of I in $\mathbb{C}[G]$ is

$$\langle \chi_I, \chi_{\mathbb{C}[G]} \rangle = \sum_{g \in G} \chi_I(g) \cdot \mathbf{1}_{g=1_G} = \chi_I(1_G) = \dim I$$

and so $\mathbb{C}[G] \simeq \bigoplus_{I \in \mathcal{I}(G)} I^{\oplus \dim I}$. We next wish to count the dimension of the centre of the endomorphism algebra of $\mathbb{C}[G]$. On one hand, we have isomorphisms of algebras[7]

$$Z\left(\mathbf{FinVect}^G(\mathbb{C}[G], \mathbb{C}[G])\right) \simeq_{\mathbf{Alg}} Z\left(\bigoplus_{I \in \mathcal{I}(G)} \mathrm{Mat}_{\dim I \times \dim I}(\mathbb{C})\right) \simeq_{\mathbf{Alg}} \bigoplus_{I \in \mathcal{I}(G)} \mathbb{C}$$

that we trust the reader[8] to construct explicitly, so $|\mathcal{I}(G)| = \dim Z\left(\mathbf{FinVect}^G(\mathbb{C}[G], \mathbb{C}[G])\right)$. For another count, we explicitly consult the structure of $\mathbb{C}[G]$. An element $f \in \mathbf{FinVect}^G(\mathbb{C}[G], \mathbb{C}[G])$ is completely determined by the value $f(1_G)$, because G-linearity specifies that $f(g) = gf(1_G)$ for $g \in G$, and G forms a basis for $\mathbb{C}[G]$. Therefore, $f \mapsto f(1_G)$ is an isomorphism of vector spaces $\mathbf{FinVect}^G(\mathbb{C}[G], \mathbb{C}[G]) \simeq_{\mathbf{Vect}} \mathbb{C}[G]$. It is actually an *anti*-isomorphism[9] of algebras because $k \circ f \mapsto k(f(1_G)) = f(1_G)k(1_G)$; fortunately, this is still enough to conclude an isomorphism[10] $Z\left(\mathbf{FinVect}^G(\mathbb{C}[G], \mathbb{C}[G])\right) \simeq_{\mathbf{Vect}} Z(\mathbb{C}[G])$.

We will be done if we count $\dim Z(\mathbb{C}[G]) = |\mathcal{CC}(G)|$. Consider the set

$$\beta := \left\{ \sum_{h \in \mathcal{CL}(g)} h \;\middle|\; \mathcal{CL}(g) \in \mathcal{CC}(G) \right\}.$$

It is easy to verify that the elements listed in β corresponding to distinct conjugacy classes are actually distinct and lie in $Z(\mathbb{C}[G])$; that is, β is linearly independent. It is also a spanning set, for given $z = \sum_{g \in G} z_g g \in Z(\mathbb{C}[G])$, we have

$$z = \sum_{\mathcal{CL}(g) \in \mathcal{CC}(G)} z_g \sum_{h \in \mathcal{CL}(g)} h.$$

Therefore, β is a basis and we conclude that $|\mathcal{I}(G)| = \dim Z(\mathbb{C}[G]) = |\mathcal{CC}(G)|$.

4.20 Proof that dimensions of irreducible modules divide order of group: (G2)

Fix[11] $I \in \mathcal{I}(G)$. Our goal is to show that $|G|/\dim I$ is an integer. Terms like these (especially division by $\dim I$) are reminiscent of character computations; let us make the connection more apparent by exploiting an "averaging trick". For ease of notation, let

$$P(g) := \sum_{h \in \mathcal{CL}(g)} h, \qquad R := \sum_{i=1}^{k} \frac{|G|}{|\mathcal{CL}(c_i)|} P(c_i) P(c_i^{-1})$$

where $g \in G$ and $\{\mathcal{CL}(c_1), \ldots, \mathcal{CL}(c_k)\}$ is an enumeration $\mathcal{CC}(G)$, so that $P(g)$ and R are elements of the group algebra $\mathbb{C}[G]$. Extend the function $\rho_I : G \to \mathrm{GL}\, I$ linearly to a function $\rho_I : \mathbb{C}[G] \to \mathbf{FinVect}(I, I)$; then, $\rho_I(P(g))$ and $\rho_I(R)$ are linear transformations $I \to I$. It is easy to verify

[7] Unital, associative $\mathbb{C}$-algebras.
[8] Lazy.
[9] I.e. swaps the order of multiplication.
[10] Isomorphism of algebras, though right now all we need is an isomorphism of vector spaces
[11] This proof is the "variant on the second proof" given by David E Speyer at StackExchange.

that each $P(g)$ is in the centre of $\mathbb{C}[G]$, so $\rho_I(P(g))$ is G-linear. Schur's lemma then asserts that $\rho_I(P(g))$ must be a scalar multiple of the identity. Taking the trace reveals that

$$\rho_I(P(g)) = \frac{|\mathcal{CL}(g)| \cdot \chi_I(g)}{\dim I} \cdot 1_I$$

and we can apply this to find that

$$\begin{aligned}
\rho_I(R) &= \sum_{i=1}^{k} \frac{|G|}{|\mathcal{CL}(c_i)|} \cdot \frac{|\mathcal{CL}(c_i)| \cdot \chi_I(c_i)}{\dim I} \cdot \frac{|\mathcal{CL}(c_i^{-1})| \cdot \chi_I(c_i^{-1})}{\dim I} \cdot 1_I \\
&= \frac{|G|^2}{(\dim I)^2} \cdot \frac{1}{|G|} \sum_{i=1}^{k} |\mathcal{CL}(c_i^{-1})| \chi_I(c_i^{-1}) \chi_I(c_i) \cdot 1_I \\
&= \frac{|G|^2}{(\dim I)^2} \cdot \frac{1}{|G|} \sum_{g \in G} \chi_I(g^{-1}) \chi_I(g) \cdot 1_I \\
&= \frac{|G|^2}{(\dim I)^2} \cdot \langle \chi_I, \chi_I \rangle \cdot 1_I \\
\rho_I(R) &= \frac{|G|^2}{(\dim I)^2} \cdot 1_I.
\end{aligned}$$

Consider now the multiplication $R \cdot - : \mathbb{C}[G] \to \mathbb{C}[G]$. Because

$$R = \sum_{i=1}^{k} \sum_{g \in \mathcal{CL}(c_i)} \sum_{h \in \mathcal{CL}(c_i^{-1})} |\mathcal{CN}(c_i)| gh,$$

it is clear that its matrix representation (with respect to the basis G of $\mathbb{C}[G]$) contains only integer entries. Its characteristic polynomial $f(\lambda)$ thus lies in $\mathbb{Z}[\lambda]$. Now, if we restrict $R \cdot -$ to any factor of I in the factorisation $\mathbb{C}[G] \simeq \bigoplus_{J \in \mathcal{J}(G)} J^{\oplus \dim J}$, it acts as $\rho_I(R)$ does. Our earlier calculation shows that $\rho_I(R)$ acts as the scalar $(|G|/\dim I)^2$, and thus this scalar is an eigenvalue of $R \cdot -$. It is hence a root of the monic integer polynomial f; applying the rational root theorem, we conclude that $|G|/\dim I$ is an integer. ∎